



TEAM 5 - Dashboard, Integration & Automation

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Table of Contents

1. Executive Summary
2. Team 5 Objective
3. Role of Team 5 in ThreatGraph
4. Integration with Other Teams
5. Team 5 Architecture Overview
6. Technologies Used
7. Complete Workflow of Team 5
8. Detailed Module Explanation
9. SIEM Integration
10. Threat Intelligence Integration
11. Dashboard and Visualization
12. Alerting and Automation Engine
13. Jira Ticketing Workflow
14. Database Design
15. API Design
16. Implementation Plan
17. Example Attack Flow
18. Expected Outcomes
19. Future Enhancements
20. Conclusion

1. Executive Summary

ThreatGraph is a Continuous Threat Exposure Management (CTEM) platform designed to continuously discover, monitor, analyze, visualize, and operationalize organizational security risks.

Within the ThreatGraph ecosystem, Team 5 acts as the final operational layer responsible for:

- Centralized dashboard visualization
- SIEM integration
- Security alerting
- Threat intelligence correlation
- Automation workflows
- Incident ticketing
- Reporting and SOC visibility

The primary goal of Team 5 is to transform raw technical outputs generated by Teams 1–4 into actionable security operations intelligence for analysts, administrators, and decision makers.



Team 5 integrates asset discovery information, monitoring events, vulnerability findings, and attack path intelligence into a unified operational platform.

2. Team 5 Objective

The objective of Team 5 is to build a centralized Security Operations and Automation platform that:

- Integrates outputs from all ThreatGraph teams
- Provides real-time dashboards and analytics
- Correlates security findings with threat intelligence feeds
- Generates alerts for critical incidents
- Automates ticket creation and incident workflows
- Provides SIEM visibility and reporting
- Helps SOC analysts prioritize remediation actions

The system is designed to reduce manual effort, improve incident response time, and provide a complete view of organizational security posture.

3. Role of Team 5 in ThreatGraph

Team 5 acts as the final operational intelligence layer within the ThreatGraph ecosystem.

Unlike other teams that focus on scanning, monitoring, risk analysis, or attack path computation, Team 5 focuses entirely on:

- Integration
- Visualization
- Automation
- Operationalization

Team 5 receives:

Team	Data Received
Team 1	Asset inventory, IPs, domains, exposed services
Team 2	Monitoring events, change detection alerts
Team 3	Vulnerabilities, CVSS, EPSS, risk scores
Team 4	Attack paths, blast radius, lateral movement

After receiving the data, Team 5:

- Normalizes the information
 - Stores security events
 - Correlates findings
 - Pushes data into SIEM platforms
 - Displays dashboards
 - Generates alerts
 - Creates Jira tickets
 - Produces reports
-



4. Integration with Other Teams

Integration with Team 1 — Attack Surface Management

Team 1 continuously discovers external assets such as:

- Domains
- Subdomains
- IP addresses
- Open ports
- Running technologies
- Internet-facing services

Team 5 receives these outputs and displays them within the Asset Dashboard.

Example:

```
{  
  "domain": "api.company.com",  
  "ip": "10.10.1.5",  
  "ports": [80,443],  
  "technology": ["nginx","nodejs"]  
}
```

This information helps security analysts understand the organization's exposed attack surface.

Integration with Team 2 — Monitoring & Change Detection

Team 2 continuously monitors assets and detects:

- New open ports
- DNS changes
- Configuration modifications
- Newly discovered assets
- Service changes

Example:

```
{  
  "event": "new_port_detected",  
  "host": "server01",  
  "port": 3389,  
  "severity": "high"  
}
```

Team 5 receives these events and:

- Generates alerts
 - Displays change notifications
 - Pushes logs to SIEM
 - Creates incident events
-



Integration with Team 3 — Vulnerability & Risk Analysis

Team 3 performs:

- CVE mapping
- Risk scoring
- KEV correlation
- EPSS analysis
- Vulnerability prioritization

Example:

```
{  
  "asset": "webserver01",  
  "cve": "CVE-2025-1234",  
  "cvss": 9.8,  
  "epss": 0.91,  
  "kev": true,  
  "risk_score": 97  
}
```

Team 5 uses this information to:

- Display vulnerability dashboards
- Prioritize remediation
- Generate critical alerts
- Create Jira tickets

Integration with Team 4 — Attack Path Visualization

Team 4 computes:

- Attack paths
- Lateral movement chains
- Blast radius analysis
- Privilege escalation routes

Example:

```
{  
  "source": "webserver01",  
  "target": "domain-controller",  
  "path": [  
    "webserver01",  
    "db01",  
    "dc01"  
  ],  
  "risk": "critical"  
}
```

Team 5 visualizes these attack paths and generates alerts for high-risk attack chains.



5. Team 5 Architecture Overview

The Team 5 architecture consists of multiple interconnected modules.

Teams 1–4



API Integration Layer



Data Normalization Engine



Threat Intelligence Enrichment



SIEM Storage Layer



Dashboard Visualization Layer



Alerting & Automation Engine



Jira Ticketing System



Reports & Notifications

The architecture is designed to provide centralized visibility and automated incident handling.

6. Technologies Used

Component	Technology
Backend API	FastAPI
Frontend Dashboard	React
Database	PostgreSQL
SIEM Platform	Elastic Stack
Enterprise Log Analysis	Splunk
Threat Intelligence	MISP
Reputation Analysis	VirusTotal
Ticketing System	Jira
Visualization	Kibana, Grafana
Queue/Streaming	Kafka / RabbitMQ / NATS

7. Complete Workflow of Team 5

- **Step 1 — Data Collection**

Team 5 receives data from Teams 1–4 through APIs or event queues.

Inputs include:

- Asset inventory



- Monitoring alerts
 - Vulnerability findings
 - Attack path intelligence
-

Step 2 — Data Normalization

Different teams generate different JSON structures.

Team 5 converts all data into a unified schema.

Example:

```
{  
  "asset_id": "srv-01",  
  "event_type": "vulnerability",  
  "severity": "critical",  
  "source_team": "team3"  
}
```

This standardization simplifies dashboard visualization and SIEM processing.

Step 3 — Threat Intelligence Enrichment

The system queries:

- MISP
- VirusTotal

for:

- malicious IPs
- suspicious domains
- IOC correlation
- malware reputation

This enriches security events with additional threat context.

Step 4 — SIEM Integration

Security logs and events are pushed into:

- Elasticsearch
- Splunk

This enables:

- centralized log analysis
 - security correlation
 - real-time detection
 - SOC monitoring
-

Step 5 — Dashboard Visualization

The processed data is displayed on centralized dashboards.

Dashboards include:

- Asset Dashboard



- Vulnerability Dashboard
- Monitoring Dashboard
- Threat Intelligence Dashboard
- Attack Path Dashboard

Step 6 — Alerting and Automation

Critical findings trigger:

- dashboard alerts
- SIEM alerts
- email notifications
- Jira tickets

This reduces manual effort and accelerates incident response.

8. Detailed Module Explanation

8.1 API Integration Layer

The API Integration Layer acts as the central communication gateway.

Responsibilities:

- Receive JSON payloads
- Validate requests
- Forward events to backend services
- Trigger automation workflows

Example APIs:

POST /api/team1/assets

POST /api/team2/events

POST /api/team3/findings

POST /api/team4/paths

FastAPI was selected because it provides:

- asynchronous performance
 - automatic API documentation
 - lightweight architecture
 - scalability
-

8.2 Data Storage Layer

The storage layer stores all operational security data.

- **PostgreSQL Stores:**
 - assets
 - vulnerabilities
 - incidents
 - alerts
 - tickets
- **Elasticsearch Stores:**



- logs
- SIEM events
- searchable security data

This combination supports both structured queries and fast security event searching.

8.3 Dashboard Visualization Layer

The Dashboard Layer provides centralized visibility for SOC analysts.

- **Asset Dashboard**

Displays:

- total assets
- exposed services
- internet-facing hosts
- cloud infrastructure

- **Vulnerability Dashboard**

Displays:

- CVSS scores
- EPSS scores
- KEV vulnerabilities
- critical assets

- **Monitoring Dashboard**

Displays:

- newly opened ports
- DNS changes
- infrastructure modifications
- **Attack Path Dashboard**

Displays:

- lateral movement chains
- blast radius
- privilege escalation routes
- **Threat Intelligence Dashboard**

Displays:

- malicious IPs
- IOC matches
- VirusTotal detections
- threat actor indicators

React, Grafana, and Kibana are used for frontend visualization.

8.4 Threat Intelligence Integration

Threat intelligence enrichment is performed using:

- **MISP**

MISP provides:



- IOC feeds
- malicious IPs
- malware indicators
- threat actor intelligence
- **VirusTotal**

VirusTotal provides:

- file reputation
- domain reputation
- URL analysis
- malware detections

Example:

If a suspicious IP is detected:

185.220.101.1

VirusTotal may return:

Detected by 32 security vendors

The dashboard then flags the asset as malicious.

8.5 SIEM Integration

Team 5 integrates with SIEM platforms to centralize security monitoring.

- **Elastic Stack Components**
- **Elasticsearch**

Stores logs and security events.

- **Logstash**

Processes and forwards events.

- **Kibana**

Visualizes dashboards and alerts.

- **Splunk**

Splunk provides:

- security analytics
- correlation searches
- threat hunting
- incident investigation

Example Splunk Query:

index=threatgraph severity=critical

This query displays all critical incidents.

8.6 Alerting Engine

The Alerting Engine generates real-time security notifications.

Alerts are triggered for:

- critical vulnerabilities
- exposed RDP ports



- attack paths to crown jewels
- malicious IOC matches
- high-risk infrastructure changes

Example Alert:

CRITICAL ALERT:

Attack path detected from public web server to domain controller.

Notifications can be sent through:

- Email
- Slack
- SIEM dashboards
- Jira tickets

8.7 Jira Automation

Team 5 automates ticket generation using Jira.

Automatic ticket creation conditions:

- CVSS > 9
- KEV listed vulnerabilities
- Internet-facing exposure
- Critical attack path detection

Example Ticket:

```
{
  "title": "Critical Internet Facing Vulnerability",
  "priority": "P1",
  "assignee": "SOC Team"
}
```

This helps automate remediation tracking.

9. Database Design

- **Assets Table**

id | hostname | ip | criticality

- **Vulnerabilities Table**

asset_id | cve | cvss | epss

- **Alerts Table**

id | severity | description | timestamp

- **Attack Paths Table**

source | target | path_risk

10. API Design

- **Data Ingestion APIs**



POST /api/assets

POST /api/events

POST /api/vulnerabilities

POST /api/attack-paths

- **Dashboard APIs**

GET /dashboard/summary

GET /dashboard/critical-assets

GET /dashboard/alerts

- **Automation APIs**

POST /jira/create-ticket

POST /alerts/send

11. Implementation Plan

- **Phase 1 — Infrastructure Setup**

Install and configure:

- Elasticsearch
 - Kibana
 - PostgreSQL
 - FastAPI
 - React
 - Jira
 - MISP
-

- **Phase 2 — API Development**

Develop:

- ingestion APIs
 - validation logic
 - normalization engine
-

- **Phase 3 — Dashboard Development**

Build:

- asset dashboards
 - vulnerability views
 - attack path visualization
 - monitoring panels
-

- **Phase 4 — Threat Intelligence Integration**

Integrate:

- VirusTotal APIs
 - MISP feeds
-



- **Phase 5 — Automation & Alerting**

Implement:

- Jira automation
 - Email alerts
 - SIEM correlation
-

12. Example Attack Flow

- **Step 1 — Team 1**

Discovers:

Public web server

- **Step 2 — Team 2**

Detects:

RDP Port Opened

- **Step 3 — Team 3**

Finds:

Critical CVE

CVSS 9.8

- **Step 4 — Team 4**

Computes:

Attack path to domain controller

- **Step 5 — Team 5**

Performs:

- Dashboard alert generation
- SIEM correlation
- Jira ticket creation
- SOC notification
- Incident reporting

Final Alert:

CRITICAL INCIDENT:

Public RDP Exposure + Critical Vulnerability + Attack Path Detected

13. Expected Outcomes

The platform is expected to:

- Provide centralized SOC visibility
 - Reduce incident response time
 - Improve remediation prioritization
 - Automate security workflows
 - Correlate threat intelligence
 - Simplify operational monitoring
 - Enable real-time security analytics
-



14. Future Enhancements

Future improvements may include:

- AI-driven alert prioritization
- SOAR platform integration
- Automated remediation workflows
- Cloud-native Kubernetes deployment
- Machine learning threat prediction
- Real-time anomaly detection

15. Conclusion

Team 5 serves as the operational intelligence layer of the ThreatGraph ecosystem.

The module integrates outputs from all upstream teams and transforms them into actionable security operations intelligence.

By combining:

- centralized dashboards
- SIEM integration
- threat intelligence correlation
- automated ticketing
- alert management
- operational reporting

Team 5 enables organizations to monitor, prioritize, and respond to cyber threats efficiently.

The platform provides complete visibility across the attack surface while improving automation, scalability, and incident response capabilities.